ASTANA IT UNIVERSITY · SE-2405

# Development of a Toolkit for Automated Comparison of ANSI X12 EDI Implementation Guides with an Investigation into Partner-Specific Implementation Divergence in Freight Logistics

First pre-defense report

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Supervisor position: _____

Submission deadline: 24 October 2026, 18:00 UTC+05:00

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Project proposal, requirements, architecture, prototype and evaluation plan.

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Source references and editable diagrams accompany the report.

# 1 Project proposal

## Project identity and team

**Title:** Development of a Toolkit for Automated Comparison of ANSI X12 EDI Implementation Guides with an Investigation into Partner-Specific Implementation Divergence in Freight Logistics.

**Institution:** Astana IT University. **Group:** SE-2405. **Stage:** first pre-defense. **Deadline:** 24 October 2026, 18:00 UTC+05:00.

**Supervisor:** Айтмұханбетова Эльвира Айтмұханбетқызы. **Position:** _____

| Team member | Diploma responsibility |
|---|---|
| Nurassyl Orazbek (nurashi) | Backend, comparison rules and evaluation |
| Asylan Usen (Asylann) | Review interface, corpus work, evaluation and pre-defense documentation |
| Anuar Zhenissov (azhenissov) | PDF extraction and normalization development |

Component ownership guides implementation and technical review. The repository records the work already delivered: Nurassyl developed the schemas, reference corpus, comparison CLI and initial PDF extractor; Asylan prepared the report, diagrams, review prototype and integration checks. Further contributions and reviews are recorded under each student's account.

## Problem and users

X12 214 conveys freight shipment status [P01]. Partners publish implementation guides that specify required fields, allowed codes, repetition limits and conditional rules [P02]. Comparing two PDFs requires an engineer to find corresponding requirements before interpreting their differences. Repeated segments, qualifiers and inconsistent notation make text-only comparison insufficient.

The intended users are EDI integration engineers and logistics IT analysts. They need a comparison that identifies the affected requirement, shows both constraints and provides the source pages needed to verify a finding. Their workflow priorities will be validated through the discussions described in section 2.

## Aim and objectives

The aim is to build a reproducible, evidence-linked comparison toolkit and investigate the variation found in comparable freight implementation guides.

1. Represent guide identity, structural keys, constraints and PDF evidence in a common schema.
2. Maintain a reference corpus with documented scope and revision identity.
3. Extract candidate requirements from supported PDF layouts and flag unresolved notes.
4. Align requirements by structural key and, where appropriate, qualifier.
5. Compare usage, code lists, repetition and conditions, retaining uncertainty.

6. Support source inspection, separate review decisions and reusable exports.
7. Evaluate extraction, alignment and comparison against reviewed reference data.

## Scope and current results

The first scope is English-language truckload **X12 214 / 004010**. The manually normalized reference covers B10, header L11 and the status-loop segments defined in the three selected guides. Envelopes, the party loop, MAN and MS3 are outside this pilot reference.

The implementation includes two PDF-layout parsers, source-integrity checks, exact and qualifier-assisted alignment, four-category comparison, JSON output and a local review interface. The reference contains **118 requirements** from C.H. Robinson, KBX and Domtar. Comparison produces **188 findings** with source-page references. Section 9 reports the measured regression checks and their scope.

The next application stage adds shared persistence, comparison API routes, an integrated frontend and broader document coverage. Transaction sets 204 and 210 require their own data and evaluation cases.

## Research questions and expected outcomes

**RQ1:** Which constraint categories differ in the selected guides after structural alignment? **RQ2:** Which extraction, alignment and interpretation errors limit reliable comparison? **RQ3:** How accurately and efficiently can users review source-linked findings compared with manual PDF inspection?

The expected outcomes are a versioned toolkit, a source-linked corpus, documented comparison rules, reviewed evaluation data and an account of observed guide divergence. Independent reference annotation and user task measurements are planned.

The engineering complexity lies in preserving identity across repeated segments, resolving code-list references, separating segment limits from loop limits and handling conditional language. The research evaluates these decisions rather than treating textual similarity as requirement equivalence.

# 2 Evidence of relevance

## Evidence available now

Official X12 documentation establishes the business role of shipment-status messages [P01]. Stedi's implementation-guide documentation describes partner-specific constraints and the distinction between static guides and machine-readable specifications [P02]. These establish context; they are not interviews or independent proof of demand for this particular toolkit.

The pilot corpus offers inspectable examples of the comparison problem. The examples below come from the existing normalized records, whose original PDF URLs, hashes and physical page numbers are preserved in `corpus/214-004010/guides/`.

| Observation | Document evidence | Implication for a proposed requirement |
|---|---|---|
| Usage differs even within the same transaction/version | AT7: C.H. Robinson pages 3 and 21 and KBX pages 6 and 18 mark it required; Domtar page 17 marks it optional | FR-04 must compare declared usage while retaining base-standard and partner labels |
| Explicit time-code lists differ | AT707: C.H. Robinson page 22 and KBX page 20 list LT; Domtar page 20 lists CT, ET, MT and PT | FR-05 must compare code sets with page evidence |
| A guide can contradict its own examples | Domtar page 9 uses LT in an example, but page 20's structured list excludes it | FR-07/08 retain evidence and review notes; the research codebook's G5 governs contradiction handling |
| Similar condition wording can reverse an implication | C.H. Robinson page 22: AT707 implies AT706; KBX page 18: AT706 implies AT707 | FR-06 must preserve direction, rather than equating similar words |
| Repetition is a separate property | LX loop repeat: C.H. Robinson pages 3 and 19 and KBX page 6 state 1 (KBX page 2 records its correction from 99999 to 1); Domtar page 8 states 999999, while Domtar's LX segment max-use on page 17 is 1 | FR-05 must keep max-use and loop-repeat distinct |

These observations concern the selected revisions and pilot scope. They motivate source-linked comparison and review, while operational frequency, impact and time savings require user evidence.

## Corpus sources

| Guide | Source PDF | Pages | Pilot requirements | Source SHA-256 prefix |
|---|---|---|---|---|
| C.H. Robinson | Published guide | 32 | 45 | 0d28243812f1 |

| Guide | Source PDF | Pages | Pilot requirements | Source SHA-256 prefix |
|---|---|---|---|---|
| KBX Technology Solutions | Published guide | 30 | 43 | c5e6ed5f63fa |
| Domtar Industries | Published guide | 23 | 30 | 098b98e96d79 |

Full hashes, revision notes and evidence excerpts are retained in the guide JSON records.

## Findings versus assumptions

**Document findings:** the selected guides use different notations and state different constraints; some evidence needs interpretation. **User questions:** frequency of comparison, costly errors, useful interface actions and acceptable review effort.

The three-guide selection is purposive. Revision age and authoring conventions may affect observed guide divergence. The project studies the documents rather than inferring how partners behave in production.

## Required user or client evidence

The required 3-5 user discussions or substantive client agreement remain pending in GH-10. Corpus analysis is a separate source of relevance evidence and does not replace those records.

The protocol in `research/interview-guide.md` specifies recruitment criteria, neutral questions, consent, evidence recording and how findings will alter requirements. Each real finding must be linked to at least one requirement, design decision or explicit scope decision. Unconfirmed assumptions remain labelled until those records exist.

# 3 Literature review

## Search and selection

The review connects document extraction, structural matching and evaluation. It uses eleven scientific publications and official technical sources. Eight conference publications are from 2022-2025; Sentence-BERT, schema-matching taxonomy and Cohen's kappa are retained as foundational methods. Author manuscripts, conference records and maintainer documentation provide the source evidence.

The source register records access and version limits. Recent rechecks used author manuscripts where conference pages were inaccessible. This is a focused engineering review rather than a systematic literature survey.

## Critical synthesis and design consequences

Document recognition and comparison answer different questions. The extraction literature motivates preserving layout, tables and page identity. It does not justify inferring a partner obligation from a correctly extracted phrase. The project therefore keeps normalized requirements and evidence between extraction and comparison, with a human correction path.

Matching similarly requires more than segment names. The schema-matching taxonomy supports separating structural candidate alignment from semantic judgment [S10]. The corpus's repeated L11 definitions make that separation concrete. Semantic retrieval is a reasonable future baseline [S09], but the reversed AT706/AT707 implication illustrates why similarity cannot be accepted as equivalence.

The evaluation papers motivate testing each stage and avoiding layout leakage. Independent annotation must establish a reviewed reference before comparison accuracy is scored. Page evidence and ambiguity notes remain part of that reference. Official guide documentation explains why a partner-specific constraint model is relevant [P02], while the X12 transaction description bounds the shipment-status business context [P01].

The reviewed work supports a staged design: recover document structure, normalize keys, compare constraints and review uncertain cases. The current Poppler rules provide a simple development baseline; richer document models are justified when a held-out guide exposes failures. X12 defines the shipment-status context [P01], while partner-guide documentation explains why implementation constraints vary [P02]. Domain feedback and independent reference labels are the next evidence needed for evaluating this design.

## Structured publication analysis

The transfer limitations and project decisions in the last two columns are our assessment, not claims that the publications tested freight guides. No cited benchmark score is a measurement of this project.

| Source | Task | Method | Data | Contribution or result | Transfer limitation | Project implication |
|---|---|---|---|---|---|---|
| S01 LayoutLMv3, 2022 | General document understanding | Joint text, layout and image pretraining | Forms, receipts and document benchmarks | Unified multimodal pretraining supports several document tasks | OCR/token preparation and task adaptation remain; freight constraints were not evaluated | Preserve geometry as an extraction option; measure domain accuracy separately |
| S02 Donut, 2022 | OCR-free document understanding | Image-to-sequence structured generation | Document images and task-specific structured targets | Demonstrates an alternative to an OCR-dependent pipeline | Generated output needs verification; no X12 requirement-equivalence evidence | Treat as an extractor candidate with schema/evidence validation |
| S03 DocLayNet, 2022 | Document layout analysis | Human layout annotation, repeated annotations and detector evaluation | 80,863 pages; 11 layout classes | Layout benchmark includes repeated annotation and tests cross-layout robustness | Layout labels do not encode partner constraints; avoid page leakage across splits | Preserve whole-guide splits and measure agreement independently |
| S04 DocILE, 2023 | Field and line-item extraction | Localization, extraction and layout-aware item association evaluation | 6.7k annotated business documents; 55 field classes | Separates localization/extraction from line-item association | Mostly short invoice-like documents, unlike long implementation guides | Score extraction and association separately; examine unseen layouts |
| S05 OmniDocBench, 2025 | PDF parsing evaluation | Task- and attribute-level benchmark evaluation | Diverse PDF pages and parsing annotations | Makes quality inspectable beyond one aggregate score | Parsing quality does not measure semantic divergence detection | Report errors by stage and property category |
| S06 M6Doc, 2023 | General document layout analysis | Layout annotations and detection evaluation | 9,080 pages; 74 annotation categories | Reported TransDLANet baseline reaches 64.5% mAP | Layout diversity is not evidence of X12-specific correctness | Record layout/scan limitations rather than assuming transfer |
| S07 TableFormer, 2022 | Recover table structure and cells | Transformer-based structure identification and cell detection | Table images, structures and cell annotations | Reported TEDS: 98.5% on simple and 95% on complex tables | A correct grid does not determine whether a row states an obligation | Keep parsing and requirement interpretation as separate stages |

| Source | Task | Method | Data | Contribution or result | Transfer limitation | Project implication |
|---|---|---|---|---|---|---|
| S08 PubTables-1M, 2022 | Table detection and structure recognition | Structure canonicalization and detection models | Large scientific-article table dataset | Addresses inconsistent ground truth and oversegmentation | Scientific tables differ from logistics-guide tables | Define canonical keys before evaluating alignment |
| S09 Sentence-BERT, 2019 | Efficient sentence similarity | Siamese sentence embeddings | Sentence-pair similarity evaluation | Provides a practical retrieval mechanism | Similarity is not logical equivalence; reversed implications can look similar | Retrieval baseline only, gated by structural identity and review |
| S10 Rahm and Bernstein, 2001 | Organize schema-matching methods | Taxonomy of matching approaches | Prior methods and schema-matching examples; not a freight corpus | Distinguishes element/structure and language/constraint approaches | Older survey, not modern PDF extraction or a freight benchmark | Separate alignment from comparison and expose matching rules |
| S11 Cohen, 1960 | Measure two-coder agreement | Chance-corrected agreement for nominal judgments | Nominal categorical judgments; methodological examples | Supplies the basis of kappa for two coders | Kappa depends on marginal labels and can be undefined | Also report raw agreement, matrices, counts and uncertainty |

# 4 Existing system comparison

## Method

Four systems are compared against eight common criteria. Stedi and IBM address EDI integration, Acrobat compares documents, and Docling extracts structured document content. The matrix describes documented capabilities rather than measured product performance.

**D** means the source documents the stated capability. **NE** means it was not established from the reviewed material. A partial capability is described explicitly.

| Criterion | Stedi Guides [P02] | IBM Sterling Map Editor [P03] | Adobe Acrobat Compare [P04] | Docling [P05] |
|---|---|---|---|---|
| C1 Input | D: EDI guide definitions | D: EDI, XML, SQL and other mapping formats | D: two PDF versions | D: PDF and other document formats |
| C2 X12 structure | D: partner-specific EDI structure | D: EDI mapping support | NE: X12 requirement semantics | NE: X12 requirement schema |
| C3 Cross-document comparison | NE: arbitrary PDF-guide constraint comparison | NE: cross-guide constraint comparison | D: text/graphics differences; semantic constraints NE | NE: constraint comparison |
| C4 Source-location evidence | NE: links to original imported PDF pages | NE: original guide-page links | D: changes located in compared documents | D: item provenance and geometry; PDF-link export NE |
| C5 Human inspection | D: interactive guides and validation | D: graphical mapping client | D: side-by-side view, filters and statuses | D: structured representation; dedicated constraint-review UI NE |
| C6 Local execution | NE: complete local guide-management workflow | D: offline Windows mapping application | D: desktop comparison workflow | D: local execution |
| C7 Reusable outputs | D: machine-readable guide data | D: translation maps | D: comparison report | D: JSON/Markdown and structured document data |
| C8 Programmatic integration | D: guides used in transaction processing | D: maps included in business-process models | NE: automation API in the reviewed page | D: Python API and CLI |

## Design implications

EDI systems support structured partner constraints but the reviewed pages do not establish an arbitrary-PDF comparison workflow. Acrobat is a useful document-difference baseline; its changes still need interpretation as requirements. Docling offers extraction and provenance capabilities that could support an alternative parser.

The proposed toolkit focuses on structural identity, four constraint categories and reviewable source-linked findings. Its benefit over manual or generic document comparison must be measured with the protocol in section 9.

## Evidence limits

The Stedi, Adobe and Docling pages were read directly. Docling's document model explicitly describes provenance and item geometry; this supports the qualified C4 entry, without establishing a finished PDF-link review workflow.

IBM's indexed official 6.2.2 documentation describes Map Editor as an offline Windows application and its use in translation processes. Direct retrieval of that page returned HTTP 403, so the comparison retains this access limitation. No licensed IBM or Acrobat trial was performed.

Source URLs and verification notes are retained in the bibliography and source register.

# 5 Technology comparison and decisions

The choices below follow the requirements for source traceability, reproducibility and a maintainable implementation. Each decision compares three alternatives.

| Decision | Alternatives | Trade-off | Choice and implementation status |
|---|---|---|---|
| ADR-01 Backend platform | Go/Gin; Python/FastAPI; TypeScript/Node | Go reuses the tested comparison package; Python simplifies extractor integration; Node shares a language with the intended frontend | Retain Go/Gin. The edi CLI and health endpoint exist; comparison API routes are planned |
| ADR-02 Storage | Versioned JSON; PostgreSQL/JSONB; SQLite | Files are easy to review and reproduce. PostgreSQL supports transactions and indexed JSON [P08]. SQLite simplifies a local application | Use JSON for the reference and interchange. PostgreSQL is configured in Compose; application persistence remains to be implemented |
| ADR-03 PDF extraction | Poppler text plus layout rules; PyMuPDF/pdfplumber; Docling | Poppler's layout-preserving text supports simple rules [P09]. Coordinate and table APIs offer more control [P06,P07]. Docling provides a broader layout/table pipeline [P05] with additional dependencies | Python rules over Poppler are implemented for two layouts. Test coordinate/table alternatives when the current rules fail on new guides |
| ADR-04 Component interaction | JSON/CSV files; synchronous REST; asynchronous jobs | Files preserve inspectable inputs and outputs. REST fits interactive review. Jobs add progress and retries at the cost of more infrastructure | CLI and file interchange are implemented. REST integration follows; a queue requires a measured workload need |
| ADR-05 Interface | HTML/CSS/JavaScript; React/TypeScript; CLI and CSV only | A static interface makes the current findings easy to inspect. React supports a larger shared workspace. CLI/CSV has the least setup but weaker evidence navigation | The static prototype is implemented. React remains the planned application frontend |

## Data contracts

The guide JSON Schema defines identity, structural keys, constraints and evidence. The comparison output is the Go Report/Finding structure: each report identifies two guides, category findings and unmatched keys. A separate divergence JSON Schema is no longer part of the toolkit.

The prototype builder invokes the comparison CLI and combines its output with reference records. Its CSV retains the automated verdict, values, keys and page references; review labels and notes occupy separate fields. The research study maintains a separate sheet format and codebook.

## Alternatives to evaluate

Layout-aware and generative methods [S01,S02,S07] are candidates for broader extraction coverage. They need the same schema, page evidence and held-out-guide evaluation as deterministic rules. A more complex model is justified by a demonstrated failure pattern and improved measured results.

The current measurements concern the three development guides. Database throughput, frontend task time and performance on new layouts remain evaluation work.

# 6 Requirements specification

P0 is mandatory for the stated stage; P1 is desirable; P2 is deferred. **P** denotes this pre-defense and **D** the later diploma implementation. A planned D requirement is not counted as delivered software.

## Functional requirements

| ID | Priority / stage | Requirement and measurable acceptance criterion | Verification and current state |
|---|---|---|---|
| FR-01 | P0-P | Identify inputs: every reference guide records partner, transaction, release, revision, source URL, SHA-256 and page count | Three guide files and schema checks; implemented |
| FR-02 | P0-P | Preserve evidence: all 118 reference requirements contain page/excerpt evidence; selected findings show both sources | Source audit and browser checks; implemented |
| FR-03 | P0-P | Align keys: support exact and qualifier rules; repeated definitions retain distinct keys; report unmatched requirements | Go alignment tests and 34/27/30 qualifier-pair counts; implemented |
| FR-04 | P0-P | Compare declared usage while retaining the original designator and label | Go comparison tests and source-linked findings; implemented |
| FR-05 | P0-P | Compare enumerated/reference/all-standard code sets and distinguish max use from loop repeat | Go comparison tests and cases in the corpus; implemented |
| FR-06 | P0-P | Preserve condition uncertainty: identical normalized condition text can match; differing free-text conditions require review | Go comparison tests and explicit review verdict; implemented |
| FR-07 | P0-P | Review findings: select all three pairs, filter category/verdict, search keys/text and inspect source pages | Browser checks, including AT707; implemented |
| FR-08 | P0-P | Save a separate same/different/unknown review and note; export all 188 findings without changing machine fields | Persistence and CSV round-trip checks; implemented |
| FR-09 | P0-D | Establish independent reference labels: two compatible study sheets, frozen codebook, per-category agreement and adjudication | Separate research repository; execution pending |
| FR-10 | P0-D | Record disagreement causes: extraction, alignment, wording, codebook or slip, with an adjudicated label | Codebook procedure; application to actual annotations pending |
| FR-11 | P0-P | Extract supported PDFs: output schema-valid records with evidence; reject mismatched source hash/page count; flag unclassified notes | Eight extractor tests and three-PDF regression; implemented for two layouts |

| ID | Priority / stage | Requirement and measurable acceptance criterion | Verification and current state |
|---|---|---|---|
| FR-12 | P0-D | Persist shared work with guide revisions, identifiable reviewers and independent-coder access controls | Service, database and authorization tests; planned |
| FR-13 | P0-P | Trace each preparation task to its owner, priority, deadline, acceptance criteria and PR/deliverable | Task tracker and task-linked commits; available |
| FR-14 | P0-P | Provide all eleven documentation sections and a requirements-coverage matrix | Source/PDF checks; supplied, with user-validation and supervisor-review dependencies |
| FR-15 | P0-P | Export comparison JSON with guide IDs, keys, category, verdict, both values and pages; keep unmatched keys separate | CLI output and prototype rebuild; implemented |
| FR-16 | P2-D | Add transaction sets 204/210 only with versioned reference data and regression/evaluation cases | Extension gate; deferred |
| FR-17 | P0-D | Expose extraction and comparison through documented API routes with validated requests and explicit failures | API contract and integration tests; planned |

## Non-functional requirements

| ID | Priority / stage | Requirement and acceptance criterion | Verification |
|---|---|---|---|
| NFR-01 | P0-P | Reproducibility: the same ordered guides and CLI produce identical dataset content; rebuilding changes neither the reference nor saved study data | Byte-identical data rebuild |
| NFR-02 | P0-P | Source integrity: source hashes/page counts match; every cited excerpt fragment occurs on its page | Poppler evidence audit of reference and extracted records |
| NFR-03 | P0-P | Traceability: exported findings retain both guide IDs, both keys, category and page references | JSON/CSV checks |
| NFR-04 | P0-P | Usability: labelled controls, visible focus, literal text rendering, useful empty states and no horizontal page overflow at 390 px | Browser checks and visual inspection |
| NFR-05 | P0-P | Local review: opening the packaged prototype and saving decisions sends no guide/review content to an external server | Browser request log; PDF links open on user action |
| NFR-06 | P0-P | Maintainability: Go tests/vet, extractor tests, report tests and documented build commands pass; reference records remain versioned | Test runs and CI configuration |

| ID | Priority / stage | Requirement and acceptance criterion | Verification |
|---|---|---|---|
| NFR-07 | P0-D | Input validation: malformed schema, duplicate keys and incompatible transaction/release combinations receive explicit errors before results are accepted | Existing key checks; comprehensive request validation planned |
| NFR-08 | P1-D | Performance measurement: record stage timings for at least ten repetitions on fixed inputs/hardware and report median/p95; agree a target after baseline measurement | Planned benchmark protocol |

## Requirements provenance

FR-01/02/03/11 and NFR-01/02 follow the source-data contract. FR-04/05/06 respond to the cases in section 2. FR-07/08 express the proposed analyst workflow and require user validation. FR-09/10 follow the existing research codebook. FR-13/14 cover school traceability and documentation requirements.

Each user finding or supervisor decision will identify the affected requirement, evidence and resulting change. Code regression results establish implementation behavior; research acceptance depends on the independent evaluation in section 9.

# 7 Architecture and diagrams

The toolkit separates PDF extraction, structural alignment, constraint comparison and review. This reflects the distinction between structural and constraint-based matching [S10] and allows errors to be attributed to the stage that produced them.

## Implemented workflow

1. A guide's metadata identifies the source PDF by URL, SHA-256 and physical page count.
2. The Python extractor runs Poppler and selects one of two layout parsers. It writes guide-schema JSON with page excerpts and review flags.
3. The Go edi CLI verifies evidence, aligns keys and compares usage, codes, repetition and conditions. It writes JSON findings and separate unmatched-key lists.
4. The prototype data builder invokes edi diff for the manual reference guides. It creates a browser bundle with source records and automated findings.
5. The local interface filters those findings, displays evidence and exports separate review decisions.
6. The extraction evaluator compares candidate records with the manually normalized reference within its declared segment/position scope.

The Go/Gin API currently exposes GET /healthz. PostgreSQL appears in Compose, but the API does not yet use it. Comparison API routes, shared persistence and the React application are planned.

## Contracts and boundaries

The guide schema represents loop path, segment, position, element and qualifier, together with usage, repetition, codes, conditions, evidence and optional review reasons. The Go Report/Finding types define comparison JSON. A finding is a judgment about one property of an aligned pair; a requirement without a counterpart is listed separately.

The browser bundle includes a content digest and stable finding order for the same inputs. Its F-prefixed IDs and machine verdicts belong to the toolkit workflow. The research project's U-prefixed sheets support independent annotation and cannot be replaced by a prototype export.

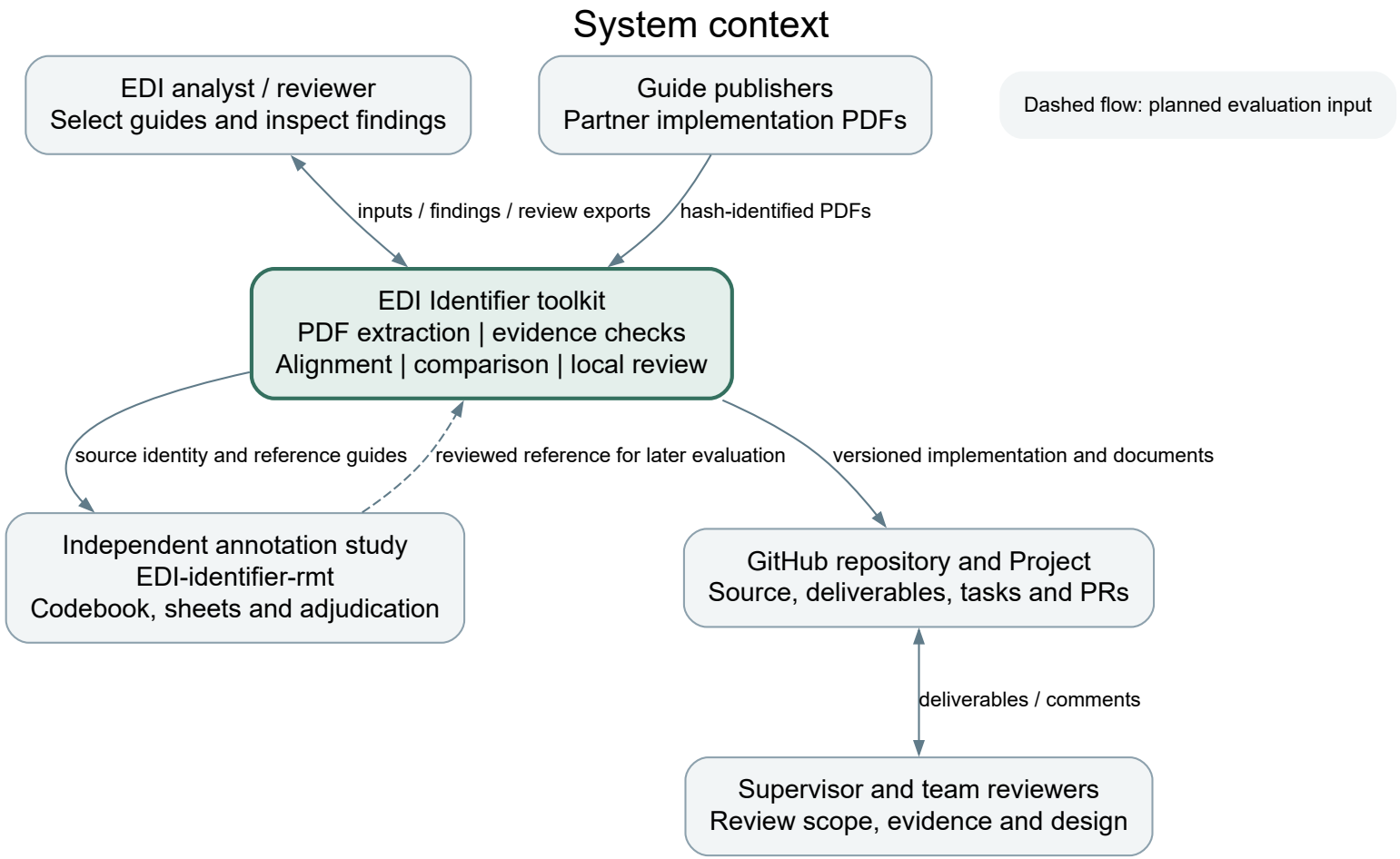
Future service routes must preserve these identities and evidence fields. Shared storage must add revision and reviewer identity, while maintaining access separation for independent coding.

## Architectural risks

Schema validity cannot establish semantic correctness. Layout rules can miss content, qualifier-assisted alignment can join different scopes and source URLs can change. Hashes, visible alignment rules, review states and an independently adjudicated reference address different parts of this risk.

## System context

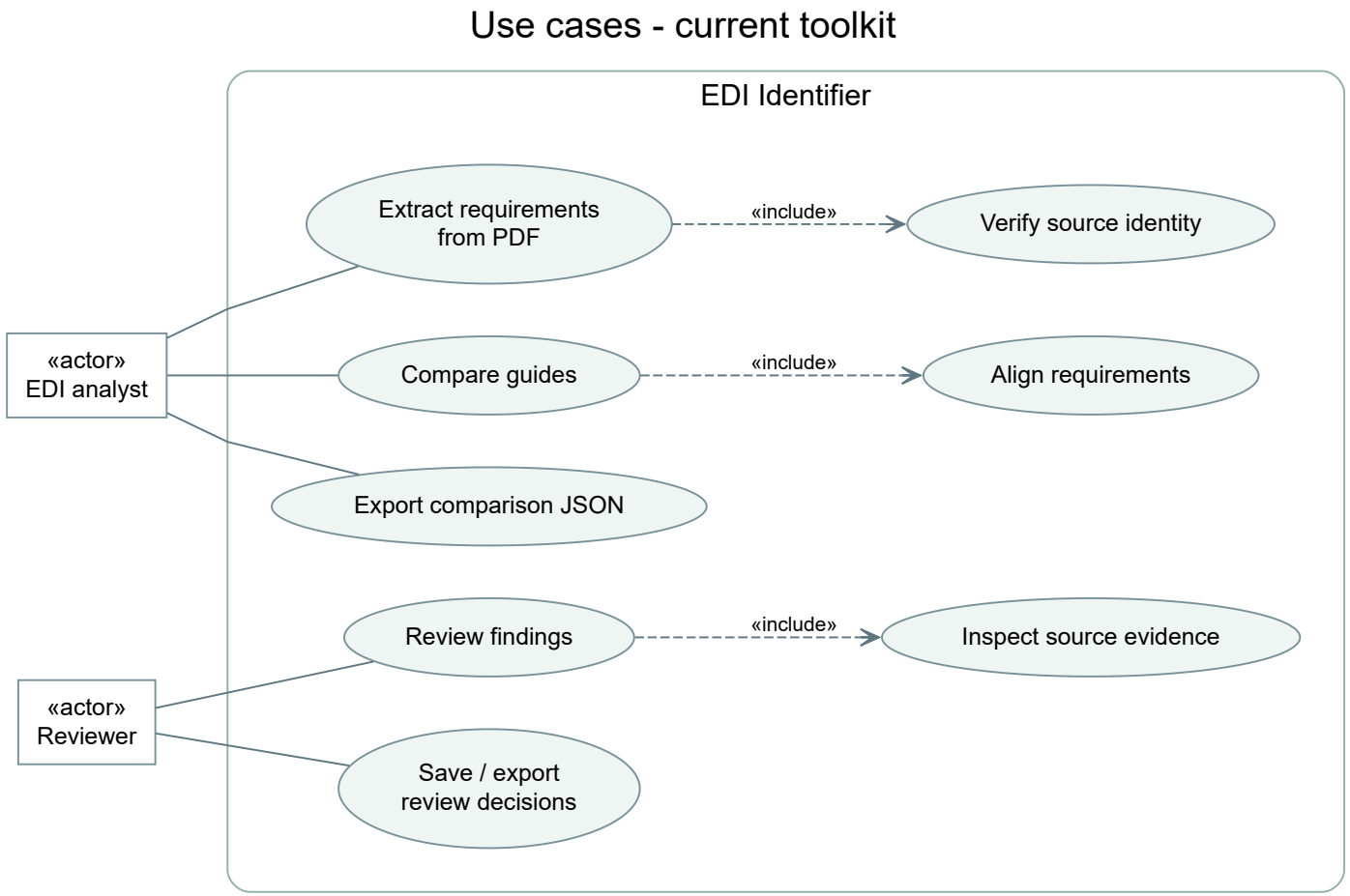
The analyst supplies comparable guides and reviews findings. Public guide providers supply PDFs. The repository and tracker hold source data, deliverables and review history. Independent annotation is maintained in the separate EDI-identifier-rmt research project; its sheets have their own IDs and protocol.




System context

## Use cases

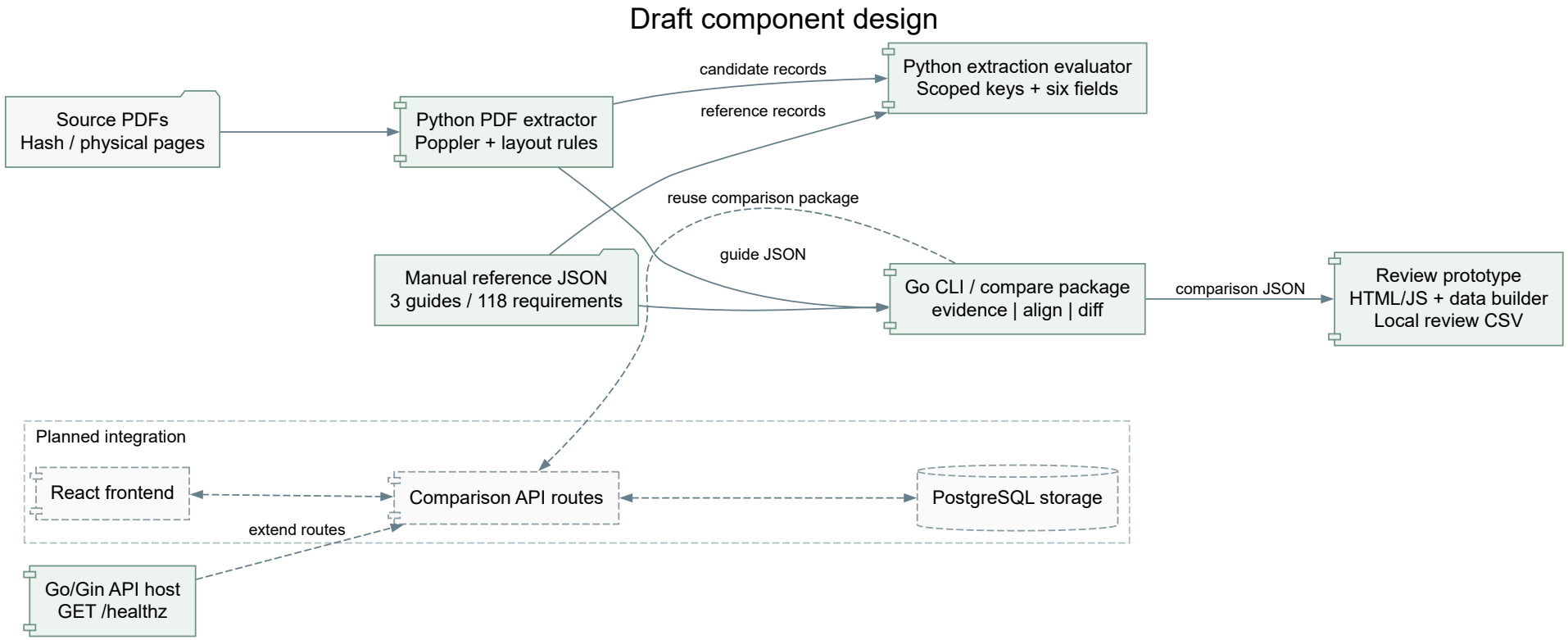
The UML use-case view covers the implemented toolkit. Actor associations are undirected; include arrows point from a use case to required supporting behavior. Extracting a PDF includes source verification; comparing guides includes structural alignment. Recording/exporting review decisions is a separate goal from exporting automatic comparison results.




Use case diagram

## Draft component design

Solid components and flows represent implemented behavior. Dashed components and flows represent the proposed service integration. The comparison package remains reusable through the CLI and the later API.




Draft component diagram

# 8 UX and prototype

The review prototype presents findings produced by the Go comparison CLI for the three manually normalized reference guides. PDF extraction is available through the separate Python command. The browser reads a generated dataset and stores review decisions locally.

## Key scenarios

1. **Select context.** Choose a guide pair, filter by category or automated verdict, and search by finding ID, key or text. The initial CHR-KBX pair contains 72 findings.
2. **Inspect a finding.** Compare both values, the alignment rule, the automated verdict and its explanation. Expand source evidence and normalized records; source links open the published PDF at the cited page.
3. **Review the result.** Save a same, different or unknown decision with a note. The decision is stored separately from the automated verdict. Reloading restores saved reviews for the same dataset.
4. **Inspect unmatched requirements.** Expand the pair's unmatched list and follow its page links. Absence of an aligned counterpart is recorded separately from a property stated in only one guide.
5. **Export reviews.** CSV export includes all 188 findings with their original values, machine verdicts, keys and page references. Only the review label and note reflect local edits.

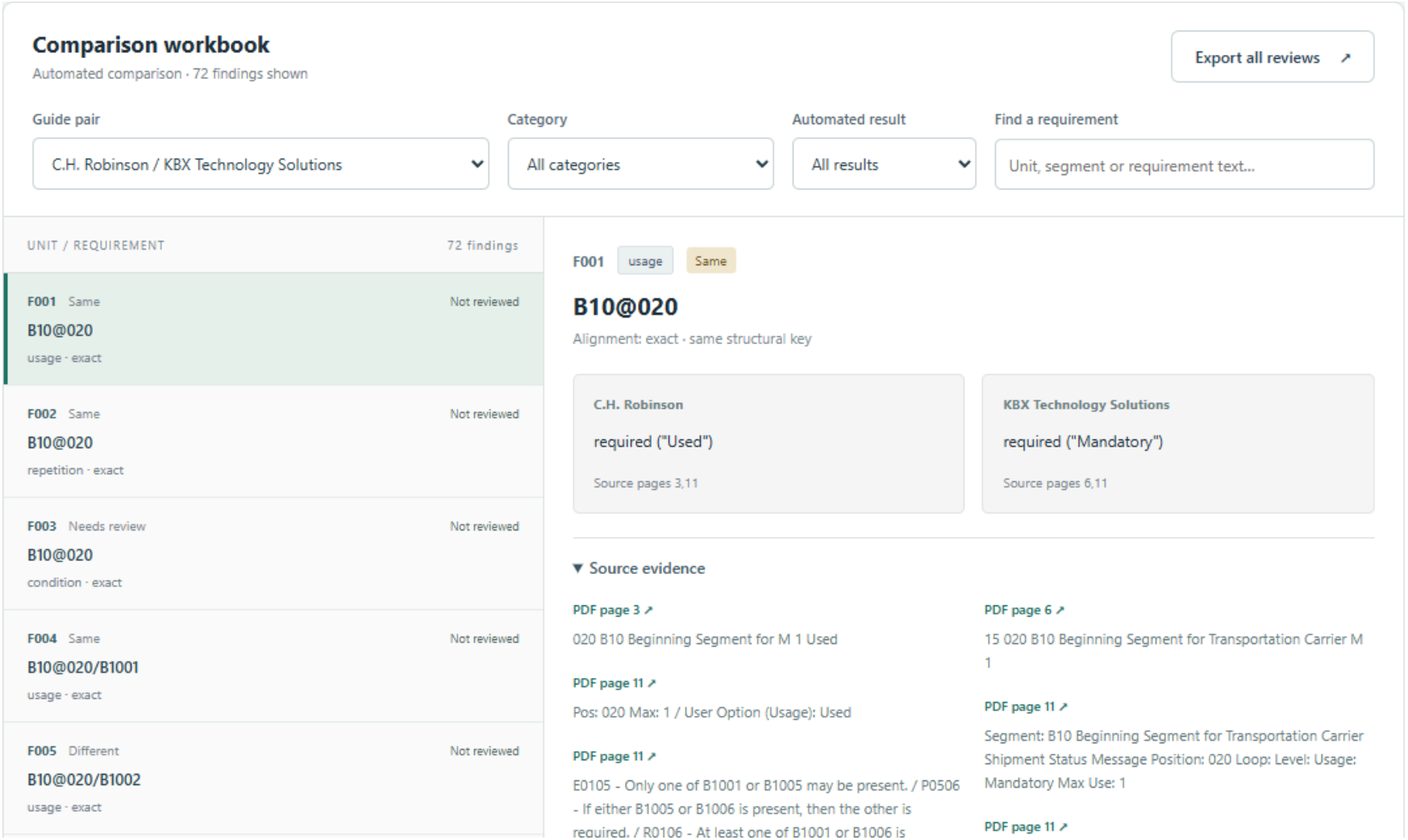
The controls have text labels, keyboard focus indicators, empty states and a message when browser storage is unavailable. The layout is checked at desktop and mobile widths. Login, shared storage, PDF upload and comparison API integration are planned components.

## Reproduction and checks

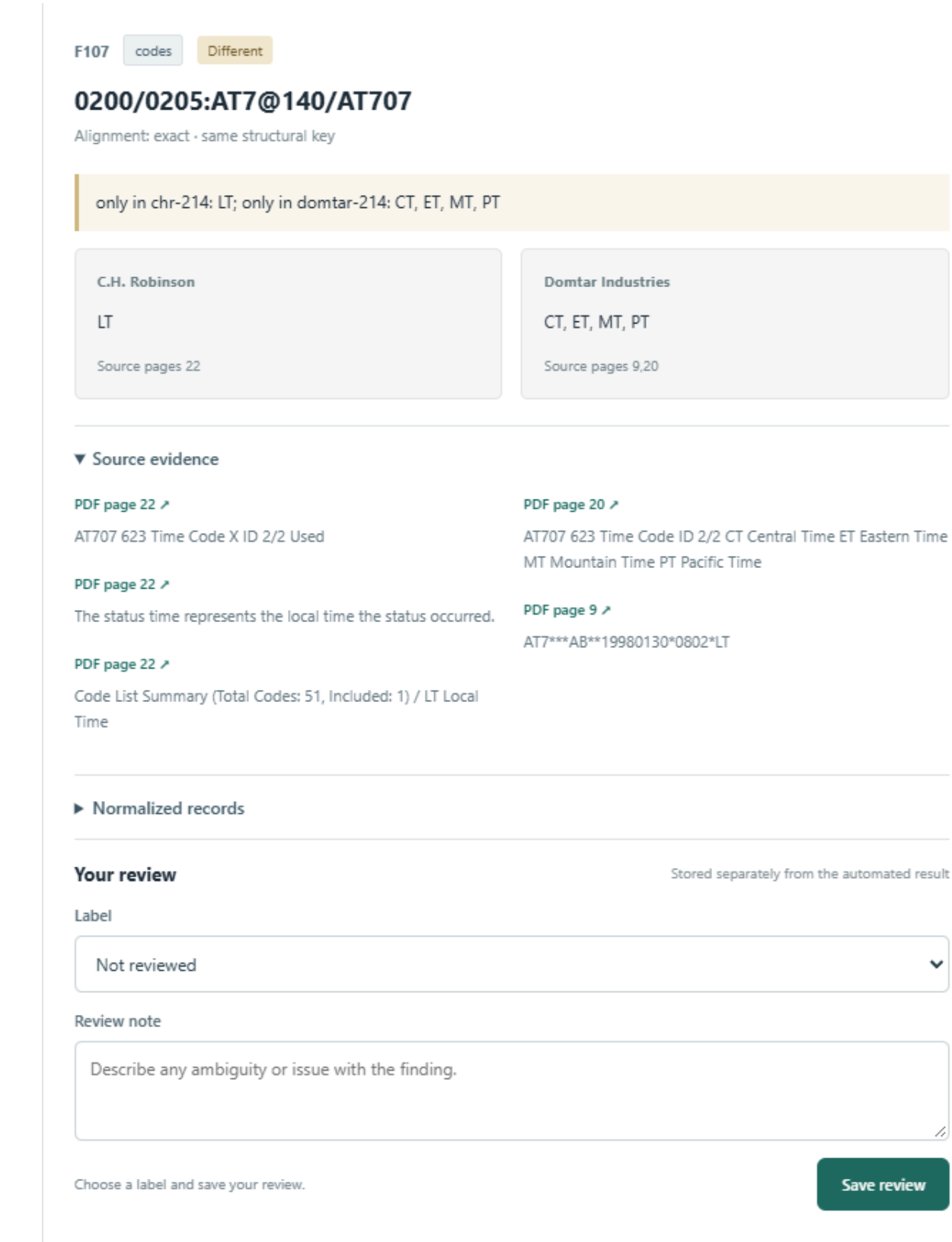
Open the prototype directly in a browser. The demonstration gives the CLI and extraction commands. The data builder compiles or uses an existing edi executable, invokes its JSON comparison report and generates the browser dataset. It retains a content digest so saved reviews are separated when inputs or findings change.

Browser checks exercise pair/category/result/search filters, evidence links, local persistence, immutable machine verdicts, empty filters, storage failure and mobile page width. A separate CSV check verifies every immutable field and quoted multiline notes. The verification record records these checks.

The independent annotation study uses the separate research repository and its own codebook and sheet format. Prototype review exports are intended for inspecting toolkit findings.




Local review prototype overview




AT707 comparison and source evidence

# 9 Verification and evaluation plan

## Measured implementation checks

The development corpus contains three manually normalized guides with 118 pilot requirements. The extractor parses wider document sections; its evaluator restricts key precision/recall to the segment/position pairs present in the pilot reference.

| Guide | All extracted records | Pilot reference keys | Matched keys / extracted in pilot scope | Six-field regression |
|---|---|---|---|---|
| C.H. Robinson | 65 | 45 | 45 / 45 | 45/45 for each field |
| KBX | 78 | 43 | 43 / 43 | 43/43 for each field |
| Domtar | 55 | 30 | 30 / 30 | 30/30 for each field |

The six fields are declared usage, base usage, max use, loop repeat, codes and syntax notes. Thus key recall/precision in this restricted reference scope is 1.000 for these development inputs. The checks do not score the 75 non-syntax reference conditions, complete-document recall or performance on unseen guides. Extraction review flags remain for 28/24/6 records respectively.

All six reference/extracted guide files pass the guide schema. PDF hashes and page counts match. Poppler 26.09.0 locates all 414 reference and 768 extracted excerpt fragments on their cited pages.

Qualifier alignment produces 34/27/30 pairs and 72/54/62 category findings. Across the 188 findings, the CLI returns 42 same, 52 different, 78 one_side and 16 review verdicts. These are automatic outputs from manually normalized inputs, not independently validated prevalence or classification accuracy.

Go tests/vet, eight extractor tests, four report-renderer tests, dataset rebuild and browser/CSV checks pass. Detailed records are in the verification record.

## Independent reference protocol

The research repository maintains the existing codebook and frozen study-sheet format. Two coders work independently against the cited PDF pages. Round 1 covers CHR-KBX units U001-U072; after both sheets are committed, disagreements are analysed and codebook revisions recorded. Round 2 uses the revised rules and is reported separately.

Disagreement causes are extraction, alignment, wording, codebook or slip. Adjudicated labels form the reference. Prototype exports use F-prefixed IDs and show automated verdicts, so they are unsuitable as independent blind-coding sheets.

Independent annotations, substantive user sessions and an adjudicated reference have not yet been supplied. Agreement, comparison F1 and time-saving results therefore remain planned.

## Metrics, baselines and measurement conditions

| Evaluation | Metric | Baseline / comparator | Conditions |
|---|---|---|---|
| Independent annotation | Raw agreement, label confusion matrix, per-category Cohen's kappa and exploratory bootstrap interval [S11] | Two independent coders; all units and both-stated subset reported separately | Frozen sheets/codebook; same IDs; label frequencies and undefined kappa reported |
| Extraction | Key precision/recall/F1, per-field accuracy, schema-valid fraction and evidence-page correctness | Current Poppler/layout rules versus a selected coordinate/table or learned parser | New complete guide/revision held out; reference labels reviewed before scoring [S03,S04] |
| Alignment | Precision/recall/F1 over adjudicated key pairs; unmatched and ambiguous-pair counts | Exact keys versus qualifier-assisted alignment | Fixed guide scope/order; many-to-one cases adjudicated |
| Constraint comparison | Per-category and macro F1, confusion matrix and coverage at each abstention level | Exact normalized-value equality; optional structurally gated embedding retrieval [S09] | Same adjudicated reference; one_side maps to unknown; review is an abstention reported separately |
| Evidence integrity | Hash/page matches and excerpt-fragment location rate | Original source PDFs and edi evidence | Frozen PDF bytes and recorded Poppler version; semantic interpretation checked separately |
| User task performance | Completion time, missed/false findings and corrections | Manual PDF comparison; optional licensed Acrobat comparison [P04] | Equal task scope, consent, separate practice, counterbalanced order and actual participant records |

Precision = TP/(TP+FP); recall = TP/(TP+FN); F1 = 2PR/(P+R). Kappa = (observed agreement - expected agreement)/(1 - expected agreement). Zero denominators and expected agreement equal to one are reported as undefined. Automation coverage and accuracy are reported together so abstention cannot inflate a quality claim.

For timing measurements, record the software version, CPU, memory, input pages/requirements and at least ten repeats; report median and p95 separately for extraction, comparison and export. A performance target follows the baseline measurement.

## Threats and decision rules

The current parsers were developed with the three evaluated guides. Their perfect pilot-field regression is therefore a development check. Hold out complete guides or revisions rather than random pages, preserve layout diversity and publish errors by stage/category [S05,S06]. Canonical structural keys reduce reference inconsistencies [S08].

Repeated requirements and guide pairs are dependent observations. Unit-level intervals are descriptive for the pilot; they do not establish population prevalence. Review actual errors before adding a model or extending transaction coverage. A new extraction method is accepted only if it improves held-out results while retaining the source-evidence contract.

# 10 Project management and risks

## Component ownership

The repository and project board hold task ownership, status and change references. GitHub Projects is the equivalent tracker permitted in the class submission instructions. Intended reviewers require access to the private board.

| Component | Responsible member | Work and review scope |
|---|---|---|
| PDF extraction (`extractor/`) | Anuar Zhenissov (@azhenissov) | Supported layouts, normalization, input validation and source evidence |
| Backend (`backend/`) | Nurassyl Orazbek (@nurashi) | Comparison rules, requirement storage, message validation and REST API |
| Frontend (`frontend/`, planned) | Asylan Usen (@Asylann) | React viewer and annotation workflow; current local prototype |
| Corpus (`corpus/`) | Asylan Usen (@Asylann) | Guide selection, versions, reference records and annotation preparation |
| Evaluation (`evaluation/`, planned) | Nurassyl Orazbek and Asylan Usen | Stage metrics, baselines, independent annotation and held-out evaluation |

Ownership identifies responsibility for development and acceptance. Commit history records actual authorship. Nurassyl supplied the initial schema, corpus, comparison and extraction foundation; Asylan prepared the report, diagrams, local review prototype and integration checks. Component-owner reviews remain open where no review has been recorded.

## First pre-defense tasks

| Task | Deliverable and acceptance | Owner / priority | Deadline and dependency |
|---|---|---|---|
| GH-2 | Assemble the reviewed package, both PDFs and submission references | Asylan / P0 | 24 Oct, 18:00; section reviews and GH-10 |
| GH-3 | Proposal, source-cited relevance and discussion protocol | Asylan / P0 | 24 Oct; real user records in GH-10 |
| GH-4 | Critical review of 20 sources; extraction-literature technical review | Anuar / P0 | 24 Oct; report draft prepared, owner review pending |
| GH-5 | Four systems/eight criteria and five technology decisions | Nurassyl, Anuar / P0 | 24 Oct; component-owner review pending |
| GH-6 | Measurable requirements and three editable architecture diagrams | Nurassyl / P0 | 24 Oct; technical review pending |

| Task | Deliverable and acceptance | Owner / priority | Deadline and dependency |
|---|---|---|---|
| GH-7 | CLI-backed prototype, evidence inspection and CSV review export | Asylan / P0 | 24 Oct; implementation checks recorded |
| GH-8 | Metrics, baselines and independent-evaluation protocol | Nurassyl, Asylan / P0 | 24 Oct; technical review pending |
| GH-9 | README, roles, milestones, risks and deliverable links | Asylan / P0 | 24 Oct; current tracker and component responsibilities |
| GH-10 | Real user/client evidence, supervisor decisions and team review | Asylan coordinates / P0 | Before submission; external participants and reviewers |
| GH-11 | Consolidated eleven-section report with bibliography and coverage | Asylan / P0 | Before submission; report sources and export checks |
| GH-13 | Readable layout, complete figures and PDF metadata | Asylan / P0 | Before submission; report and screenshot exports |

The report draft was prepared by Asylan. GH-4, GH-5, GH-6 and GH-8 were reopened for their component owners' technical review; prepared material does not establish another member's completed review. Each task links to the preparation PR and its deliverable.

## Implementation tasks and next stage

The repository also tracks the existing foundation tasks: GH-14 (schema and initial corpus), GH-15 (structural alignment), GH-18 (extraction prototype) and GH-20 (constraint comparison), completed by Nurassyl in PR #1. GH-16 remains assigned to Anuar for independent source checking of all 118 normalized records, due 23 October. Locating excerpts automatically does not complete that semantic review.

| Task | Planned deliverable | Owner | Stage / acceptance gate |
|---|---|---|---|
| GH-21 | Unseen-layout extraction and validated input contract | Anuar | After first pre-defense; dates agreed with supervisor |
| GH-22 | Requirement storage, comparison API and validation scope | Nurassyl | After first pre-defense; dates agreed with supervisor |
| GH-23 | React divergence viewer and annotation workflow | Asylan | After first pre-defense; dates agreed with supervisor |
| GH-24 | Versioned guide selection and held-out reference corpus | Asylan | After first pre-defense; dates agreed with supervisor |
| GH-25 | Independent annotation and held-out evaluation | Nurassyl, Asylan | After first pre-defense; dates agreed with supervisor |

These five P1 tasks describe subsequent implementation and evaluation. Their acceptance criteria remain unchecked. They are not required to be completed as production application features by 24 October.

## Milestones

| Milestone | Date / stage | Exit condition |
|---|---|---|
| M0 Foundation | 15-20 October | Repository/API skeleton, schema, manual corpus and initial tools in PR #1 |
| M1 Integrated prototype | 21-22 October | PDF extraction, comparison, local review and regression checks reproducible |
| M2 Pre-defense review | 22-23 October | Report/diagrams consistent; technical corrections, user evidence and supervisor responses recorded |
| M3 Submission | By 24 October, 18:00 UTC+05:00 | Both PDFs reviewed, access verified, team PR review and exact submission version identified |
| M4 Application integration | After first pre-defense | Comparison API, shared storage and frontend meet agreed acceptance tests |
| M5 Evaluation and extension | Following M4 | Independent reference, unseen-guide and user-task evaluation; scope extension decided from results |

Dates for M4/M5 depend on the supervisor's academic schedule.

## Risk register

| ID | Risk | Likelihood / impact | Mitigation and owner |
|---|---|---|---|
| K01 | User validation unavailable by the deadline | High / High | Schedule 3-5 relevant discussions or obtain a substantive client agreement; Asylan |
| K02 | Substantive supervisor review pending | High / High | Review the package against the decisions in section 11 and record responses; Asylan |
| K03 | Three development guides overstate generalization | High / High | Bound results and add unseen guides/revisions; research team |
| K04 | Source URL or PDF revision changes | Medium / High | Preserve metadata and verify hashes before accepting inputs; team |
| K05 | Extraction/alignment error mistaken for divergence | High / High | Separate stage metrics, source evidence and cause-coded error review; team |

| ID | Risk | Likelihood / impact | Mitigation and owner |
|---|---|---|---|
| K06 | Committee cannot access a private resource | Medium / High | Verify repository/board access with the intended reviewer accounts; Asylan |
| K07 | Responsibility mistaken for completed contribution | Medium / High | Record actual work and peer review under individual accounts; all members |
| K08 | Application scope delays stage deliverables | Medium / High | Keep service integration separate from the required prototype; Asylan |
| K09 | Missing Go/Poppler/browser dependencies | Medium / Medium | Document versions and commands; preserve CI checks; team |
| K10 | Submitted files differ from the identified version | Medium / High | Review both exports and create the immutable stage tag at the submitted commit; Asylan |

## Review and release

Component-owner reviews are recorded in the preparation PR and linked tasks. User validation and substantive supervisor feedback remain separate acceptance dependencies. Review ownership does not transfer authorship of existing code or report material.

The submission uses an immutable pre-defense-1 tag after the team checks both PDFs and the exact submitted commit. The form is submitted once on behalf of the team.

# 11 Supervisor feedback

**Supervisor:** Айтмұханбетова Эльвира Айтмұханбетқызы.

Substantive review of the current deliverables is pending in GH-10. Administrative acceptance of the team is separate from technical review.

## Decisions requested

| Deliverable | Review question | Expected decision record |
|---|---|---|
| Proposal and scope | Is the initial 214/004010 truckload scope appropriate for the engineering and research aims? | Accepted scope or a specific adjustment |
| Relevance and requirements | Which user/client evidence is sufficient, and which workflow assumptions need validation first? | Evidence to collect and affected requirements |
| Literature and comparisons | Are the selected methods, systems and baseline choices relevant? | Sources or alternatives to add/change |
| Architecture and prototype | Are the current CLI/extractor and planned service boundaries clear? | Component or scenario corrections |
| Evaluation | Are independent annotation, guide-level holdout and abstention metrics adequate? | Required evaluation cases and acceptance expectations |

## Comment and response register

For each received comment, record the date, reviewed version, requested change, decision, changed artifact/commit and follow-up status.

| Reviewed material | Comment / date | Response decision | Change reference | Status |
|---|---|---|---|---|
| Current project package | Awaiting substantive review | Pending | GH-10 | Open |

The register will be updated with the supervisor's actual comments and resulting revisions.

# Bibliography

Scientific core S01-S08: conference publications from 2022-2025. S09-S11 supply foundational matching and agreement methods. The source register records verification and access limits.


- **[S01]** Yupan Huang, Tengchao Lv, Lei Cui, Yutong Lu, Furu Wei. 2022. LayoutLMv3: Pre-training for Document AI with Unified Text and Image Masking. ACM Multimedia 2022. DOI: 10.1145/3503161.3548112.
- **[S02]** Geewook Kim et al. 2022. OCR-Free Document Understanding Transformer. ECCV 2022. DOI: 10.1007/978-3-031-19815-1_29.
- **[S03]** Birgit Pfitzmann, Christoph Auer, Michele Dolfi, Ahmed S. Nassar, Peter Staar. 2022. DocLayNet: A Large Human-Annotated Dataset for Document-Layout Analysis. KDD 2022. DOI: 10.1145/3534678.3539043.
- **[S04]** Štěpán Šimsa et al. 2023. DocILE Benchmark for Document Information Localization and Extraction. ICDAR 2023. DOI: 10.48550/arXiv.2302.05658.
- **[S05]** Linke Ouyang et al. 2025. OmniDocBench: Benchmarking Diverse PDF Document Parsing with Comprehensive Annotations. CVPR 2025, pp. 24838-24848. Supporting source.
- **[S06]** Hiuyi Cheng et al. 2023. M6Doc: A Large-Scale Multi-Format, Multi-Type, Multi-Layout, Multi-Language, Multi-Annotation Category Dataset for Modern Document Layout Analysis. CVPR 2023.
- **[S07]** Ahmed Nassar, Nikolaos Livathinos, Maksym Lysak, Peter Staar. 2022. TableFormer: Table Structure Understanding With Transformers. CVPR 2022, pp. 4614-4623. Supporting source.
- **[S08]** Brandon Smock, Rohith Pesala, Robin Abraham. 2022. PubTables-1M: Towards Comprehensive Table Extraction From Unstructured Documents. CVPR 2022, pp. 4634-4642.
- **[S09]** Nils Reimers, Iryna Gurevych. 2019. Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. EMNLP-IJCNLP 2019, pp. 3982-3992. DOI: 10.18653/v1/D19-1410.
- **[S10]** Erhard Rahm, Philip A. Bernstein. 2001. A Survey of Approaches to Automatic Schema Matching. The VLDB Journal 10(4).
- **[S11]** Jacob Cohen. 1960. A Coefficient of Agreement for Nominal Scales. Educational and Psychological Measurement 20(1), pp. 37-46. DOI: 10.1177/001316446002000104.
- **[P01]** X12. n.d. 214 Transportation Carrier Shipment Status Message. Official documentation.
- **[P02]** Stedi. n.d. What is an EDI guide?. Official documentation.
- **[P03]** IBM. n.d. Associated External Applications, Sterling B2B Integrator 6.2.2. Official documentation. Supporting source.
- **[P04]** Adobe. n.d. Compare two versions of a PDF file in Adobe Acrobat. Official documentation.
- **[P05]** Docling project. n.d. Docling documentation. Official documentation. Supporting source.
- **[P06]** PyMuPDF. n.d. The Basics and text extraction documentation. Official documentation.
- **[P07]** Jeremy Singer-Vine and contributors. n.d. pdfplumber documentation. Official documentation.
- **[P08]** PostgreSQL Global Development Group. n.d. JSON Types. Official documentation.
- **[P09]** Poppler contributors; Debian manpages. n.d. pdftotext(1): PDF to text converter. Official documentation.

# First pre-defense requirements matrix

Requirements source: the school's 'До 24 октября' document and class submission instructions. Deadline: 24 October 2026, 18:00 UTC+05:00.

Available means that the document or prototype is present. Review identifies a remaining consistency or acceptance check. Pending identifies missing evidence or a submission action.

| ID | Requirement | Project material | Evidence | Status | Remaining work |
|---|---|---|---|---|---|
| R01 | Proposal: title, team, supervisor, problem, users, aim, objectives, outcomes and scope | `sections/01-proposal.md`, README | Section 01: problem, users, objectives, scope and team | Review | Confirm final team details |
| R02 | 3-5 user discussions or substantive client agreement | `sections/02-relevance.md`, `research/interview-guide.md` | Discussion protocol prepared; no completed discussion records | Pending | Collect discussion or client records and link findings to requirements |
| R03 | Critical review: at least 12 meaningful sources and 8 scientific publications | `sections/03-literature.md`, `BIBLIOGRAPHY.md`, `sources.json` | 20 used sources, including 11 scientific publications | Available | Review relevance and claim-to-source links |
| R04 | 3-5 systems, 6-10 consistent comparison criteria and supporting evidence | `sections/04-systems.md` | Four systems compared against eight shared criteria | Available | Review comparison claims and documented limits |
| R05 | At least 3 technical decisions, 2-3 alternatives each | `sections/05-technologies.md` | Five decisions, three alternatives per decision | Available | Five decisions reflect the current stack; revisit after evaluation |
| R06 | Functional/non-functional requirements, priorities and acceptance criteria | `sections/06-requirements.md` | 17 functional and eight non-functional requirements with acceptance criteria | Review | Current behavior and planned requirements separated; review with supervisor |
| R07 | Context, use-case and draft component diagrams with editable sources | `sections/07-architecture.md`, `diagrams/` | Three diagrams: DOT, SVG and PNG; use cases also in PlantUML | Available | Review scope with supervisor |
| R08 | UX scenarios and reproducible interaction examples | `sections/08-prototype.md`, `prototype/`, `DEMO.md` | 188 CLI findings; desktop/mobile browser and CSV checks | Available | Review the demonstrated workflow with a prospective user |

| ID | Requirement | Project material | Evidence | Status | Remaining work |
|---|---|---|---|---|---|
| R09 | Verification plan, metrics, measurement conditions and baseline | `sections/09-evaluation.md`, `evidence/verification.md` | 198 extracted records; 118 reference keys; six-field regression; future study protocol | Review | Development regression recorded; independent and held-out evaluation pending |
| R10 | Repository, tracker, responsibilities, milestones, risks and dependencies | `sections/10-management.md`, `tracker/README.md` | 21 repository tasks across all three members; preparation board, milestones, roles and risks | Review | Confirm task links and reviewer access |
| R11 | Supervisor comments and response decisions | `sections/11-feedback.md`, GH-10 | Review questions prepared; substantive feedback pending | Pending | Record substantive comments and resulting changes |
| G01 | Repository access and identifiable individual contributors | README and repository access settings | Named contributors in Git; private-board access unconfirmed | Pending | Verify team, supervisor and committee access |
| G02 | Complete project README with setup, execution and testing instructions | Root README | Setup, extraction, comparison, prototype and test commands | Available | Setup, CLI/extraction commands and local checks verified |
| G03 | Clear source/docs/diagram/test/config structure and suitable ignore rules | Repository directories and `.gitignore` | Backend, extractor, schema, corpus, tests, docs and CI directories | Available | Complete publication and history review |
| G04 | Progressive contributions and review activity | Task-linked commits and pull requests | Task-linked changes; teammate review still required | Review | Confirm individual contribution and review records |
| G05 | Assigned tasks with outcomes, criteria, deadlines and change links | Live Issues and project board; `tracker/README.md` | 21 assigned Issues: preparation, foundation and next-stage component work | Review | Obtain component-owner reviews and verify change links |
| G06 | Task branches and peer-reviewed pull requests | PR #1 and PR #12 | PR #1 foundation and PR #12 preparation branch | Pending | Obtain team-member review of the published preparation PR |
| G07 | Immutable tag identifying the submitted version | `pre-defense-1` at the submitted commit | Submission tag has not been created | Pending | Tag the reviewed submission version |
| G08 | Demonstrate repository, task traceability, review and available implementation | `DEMO.md`, task tracker and version history | DEMO.md and recorded local verification | Review | Rehearse the current commands and selected changes |

| ID | Requirement | Project material | Evidence | Status | Remaining work |
|---|---|---|---|---|---|
| S01 | One team submission with project, supervisor, members, contribution and links | Team submission record | One team form remains to be completed and submitted | Pending | Complete and check the form fields |
| S02 | Separate PDF of board and task screenshots | Board and task screenshot PDF | Fresh board overview and individual task captures in a separate PDF | Available | Confirm reviewer access and review the snapshot before submission |
| S03 | One consolidated report PDF with all required sections | Consolidated report PDF; eleven sections, bibliography and matrix | One PDF: eleven sections, bibliography and coverage matrix | Available | Final version and submission review |

## Reference data

The implementation includes schema-branch commit `cf4d3d2e15e3c8464804f6f9255caf6795f174ad`: three manually normalized X12 214/004010 guides, 118 requirements, PDF extraction and four-category comparison. The prototype shows 188 findings generated by the current CLI. Independent annotation tooling is maintained in the separate research repository. Architecture, requirements and evaluation distinguish the implemented toolkit from planned service integration.

## Tracker interpretation

The DOCX names YouTrack. The class instructions permit an equivalent system. GitHub Projects and repository Issues provide owner, priority, deadline, acceptance criteria, status and change references. The private board still requires reviewer access verification.